

CMPSC 480 Comprehensive Final Examination

Weeks 1-15: methods, evidence, integration, and responsibility

CMPSC 480 - Student Question Deck

Assessment overview

Solve applied, mathematical, algorithmic, and engineering problems under explicit assumptions.

Coverage: All modules with emphasis on machine learning, evaluation, fairness, deployment, and cross-module integration

Points

150 total

Expected time

180 minutes

Submission

Individual examination PDF

Learning objectives

- ▶ Compare and apply classical and adversarial search.
- ▶ Reason about stochastic planning and reinforcement learning.
- ▶ Perform Bayesian exact-inference operations.
- ▶ Derive regression, classification, and backpropagation updates.
- ▶ Design valid evaluation and hyperparameter protocols.
- ▶ Analyze fairness and responsible deployment decisions.
- ▶ Integrate methods through testable engineering contracts.

Requirements checklist

- ▶ Work independently and show intermediate algebra, frontier state, or factor state.
- ▶ One double-sided handwritten letter-size reference sheet is permitted.
- ▶ A nonprogrammable calculator is permitted; networked devices and communication are prohibited.
- ▶ State every tie-breaking, terminal, reward, and zero-division convention you use.
- ▶ Answers receive credit for justified reasoning; an unsupported final number may receive little credit.
- ▶ Academic integrity rules of the course and institution apply throughout the examination.

Task 1 - Search and heuristic guarantees (15 points)

A* runs on a positive-cost graph with a heuristic that is admissible but not consistent.

- a. State admissibility and consistency formally. [5 points]
- b. Can graph-search A* remain optimal with this heuristic? State the needed implementation behavior. [5 points]
- c. Design a test that exposes missing reopening. [5 points]

Task 2 - Adversarial reasoning (15 points)

A stochastic game node chooses action A with outcomes 8 and 0 at probabilities 0.25 and 0.75, or action B with guaranteed value 3.

- a. Compute expectimax values and choose an action. [5 points]
- b. What would a worst-case MIN interpretation choose for A, and why is that a different model? [5 points]
- c. Give two tests for a combined minimax-expectimax implementation. [5 points]

Task 3 - MDPs and reinforcement learning (15 points)

For $Q(s,a)=2$, $\text{reward}=-1$, $\gamma=0.9$, $\max_{a'} Q(s,a')=5$, and $\alpha=0.1$, compare one Q-learning update with a Bellman optimality backup.

- a. Compute the Q-learning target, TD error, and updated Q. [5 points]
- b. How would the terminal case differ? [5 points]
- c. Compare information required by value iteration and Q-learning. [5 points]

Task 4 - Bayesian inference (15 points)

A disease prior is 0.02, test sensitivity is 0.9, and false-positive rate is 0.1.

- a. Compute $P(\text{disease}|\text{positive})$. [5 points]
- b. Explain how variable elimination avoids repeated enumeration. [5 points]
- c. Compare rejection sampling with likelihood weighting for rare evidence. [5 points]

Task 5 - Regression and gradient descent (15 points)

$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $y = [1, 3, 5]$, and $w = [0, 0]$ for one-half mean squared error.

- a. Compute the initial residual and gradient. [5 points]
- b. With $\alpha = 0.1$, compute the new weights. [5 points]
- c. Explain how feature scaling and collinearity affect optimization or solving. [5 points]

Task 6 - Neural networks and backpropagation (15 points)

A layer has A_{prev} shape (4,20), W shape (3,4), b shape (3,1), and delta shape (3,20).

- a. Give Z , A , dW , and db shapes. [5 points]
- b. Write the previous-layer delta formula and shape. [5 points]
- c. Describe a two-sided gradient check and one limitation. [5 points]

Task 7 - Classification and regularization (15 points)

A binary classifier has logit $z=\log(4)$, label $y=1$, $w=[3,4]$, $\lambda=0.5$, and $m=5$.

- a. Compute $\text{sigmoid}(z)$ and cross-entropy loss. [5 points]
- b. Compute the L2 penalty and gradient addition. [5 points]
- c. Explain why threshold 0.5 is not universally optimal. [5 points]

Task 8 - Evaluation and tuning (15 points)

A rare-event classifier yields $TP=12$, $FP=8$, $FN=3$, and $TN=977$.

- a. Compute accuracy, precision, recall, and F1. [5 points]
- b. Training and validation error are both high and close. Diagnose and name one experiment. [5 points]
- c. Give a valid hyperparameter workflow with k-fold CV. [5 points]

Task 9 - Fairness and responsible deployment (15 points)

Group A0 has $\text{TPR}=0.85$ and $\text{FPR}=0.10$; group A1 has $\text{TPR}=0.65$ and $\text{FPR}=0.25$. A production feature distribution then shifts.

- a. Compute the equalized-odds gaps and interpret them. [5 points]
- b. Explain why removing the protected attribute alone does not establish fairness. [5 points]
- c. Specify a complete shift monitor and response. [5 points]

Task 10 - Cross-module integration (15 points)

Design a system that combines a learned heuristic, Bayesian belief state, and a search or control policy for navigation under uncertain hazards.

- a. Define the component handoffs. [5 points]
- b. State three component or end-to-end tests. [5 points]
- c. Design a baseline, metrics, and limitation statement. [5 points]

Edge cases and defensive programming

- ▶ A logistic probability is numerically zero.
- ▶ A subgroup metric denominator is zero.
- ▶ A ReLU check lands at zero.
- ▶ A learned heuristic overestimates.
- ▶ A production dependency fails during evaluation.

Submission instructions

- ▶ Write your name and identify every requested problem and subpart.
- ▶ Show enough intermediate work that another reader can audit the result.
- ▶ Submit the complete question-response document when time is called.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric - part 1

Category	Points	Required evidence
Search and heuristic guarantees	15	Guarantee analysis, implementation rule, and counterexample reasoning.
Adversarial reasoning	15	Expectimax calculation, minimax contrast, and engineering test.
MDPs and reinforcement learning	15	Numeric update and planning-learning comparison.
Bayesian inference	15	Posterior, factor reasoning, and approximate-inference choice.
Regression and gradient descent	15	Gradient calculation, update, and conditioning analysis.

Grading rubric - part 2

Category	Points	Required evidence
Neural networks and backpropagation	15	Shape derivation, gradient formulas, and checking.
Classification and regularization	15	Probability, loss, L2 penalty, and threshold interpretation.
Evaluation and tuning	15	Metrics, curve diagnosis, and selection protocol.
Fairness and responsible deployment	15	Disparity, criterion reasoning, and monitoring control.
Cross-module integration	15	Architecture, verification, and responsible evidence plan.

Reflection prompt

In one or two sentences:

Name one connection among modules that changed how you would design or evaluate an AI system.